BIG DATA WITH PYTHON · SESSION 3

Cleaning and Visualizing Data

Find the problems in a dataset, fix them, then use plots to see what the numbers are saying.

Notebook: BigData-session-3.ipynb · Data: sherlock/sherlock_mystery_2apps.csv

A BIGGER DATASET

Always start with a first look

```
df2 = pandas.read_csv('sherlock/sherlock_mystery_2apps.csv') # about 76 MB
df2_backup = df2.copy() # a safety copy
```

How big?
df2.shape

What types?
df2.info()

Typical values?
df2.describe().T

What's inside?
df2.head()

WHY CLEANING MATTERS

Up to two-thirds of the time goes into preparing the data



Today: steps 1 and 2 · Step 3 comes in the machine learning module

WHAT CAN GO WRONG

Four common problems in real data

Missing

Empty cells, shown as NaN

Irrelevant

Columns that tell us nothing

Duplicate

The same rows or columns twice

Bad or inconsistent

Impossible values, mixed spellings

In df2 we'll find **1 irrelevant column**, **1 column with missing values** and **2 duplicated columns**.

KNOW YOUR DATA

What kind of values does each column hold?

Categorical

Labels or groups. Math on them makes no sense.

`ApplicationName`

Numerical · continuous

Any value in a range, like a measurement.

`CPU_USAGE`, `vsize`

Numerical · discrete

Whole-number counts only.

`num_threads`

> **Numbers aren’t always quantities.** priority holds only a few whole numbers, so it acts more like a ranked category.

LOOK BEFORE YOU CLEAN

Plot every column with a loop

```
fig = pyplot.figure(figsize=(16.0, 10.0))
columns = (c for c in df2.columns if c != 'ApplicationName')
for i, col in enumerate(columns):
    axes = fig.add_subplot(5, 3, i + 1) # 5 rows × 3 columns of small plots
    df2[col].plot(ax=axes) # or: seaborn.boxplot(x=df2[col], ax=axes)
```

Look for one column that behaves unlike all the others.

Look for two plots that look exactly the same.

CLEAN 1 · IRRELEVANT DATA

Drop a column that only repeats the row number

row label	Unnamed: 0
0	0
1	1
2	2
…	…

Same values on every row

```
# prove it: True means every row matches
(df2['Unnamed: 0'] == df2.index).all()

# then remove it
df2 = df2.drop(columns=['Unnamed: 0'])
```

It tells us nothing new, so it only gets in the way.

Find the empty cells

A	B	C	D
1	2	3	0
3	4	NaN	1
NaN	NaN	NaN	NaN
NaN	3	NaN	4

The toy table ex0 from the notebook. NaN = missing

```
ex0.isna() # True where missing
ex0.isna().sum() # count per column

df2.isna().sum() # now on real data
```

sum() works because **True counts as 1** and False as 0.

CLEAN 2 · MISSING DATA

Three ways to handle missing values

Option	Code	Good	Watch out
Drop rows	df2.dropna()	Keeps only real values	Loses rows
Drop the column	df2.drop(columns=['cminflt'])	Keeps every row	Loses a feature
Fill in a value	df2.fillna(value)	Keeps everything	Guesses can mislead

Our choice: drop the rows. About 22% of rows are missing cminflt, but over 600,000 rows remain.

CLEAN 3 · DUPLICATE DATA

Spot rows or columns that appear twice

Duplicate rows

```
df3.duplicated()
df3.drop_duplicates()
```

duplicated() is True for each repeat. drop_duplicates() removes them.

Duplicate columns

```
(df2['vsize'] == df2['Mem']).all()
```

True means the two columns are identical on every row. Keep one.

> **Check before you delete.** Two different measurements can, rarely, have the same values by chance.

VISUALIZATION

Pick the plot that fits your question

Your question	Plot	Function
How many rows in each category?	Count plot	seaborn.countplot
How are the values spread out?	Histogram	pyplot.hist · seaborn.histplot
Where are the median, quartiles, outliers?	Box plot	seaborn.boxplot
What is the average per category?	Bar plot	seaborn.barplot
Do two columns move together?	Scatter plot	seaborn.scatterplot
Which pairs of columns are related?	Heat map	seaborn.heatmap

HISTOGRAM

How are the values spread out?

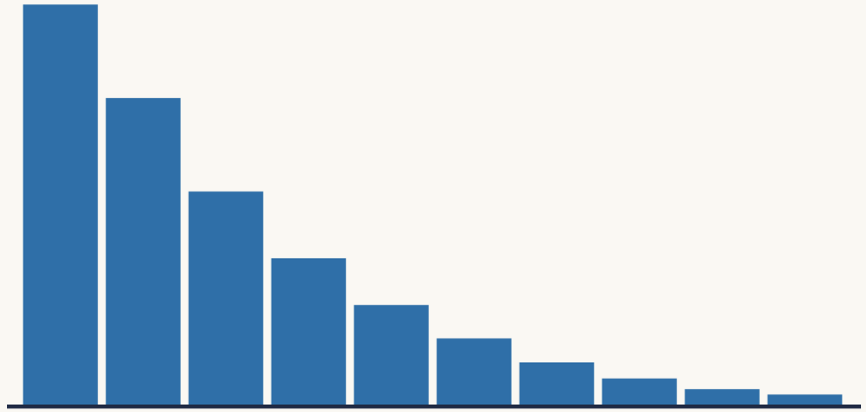


Illustration only: many small values, a few large ones

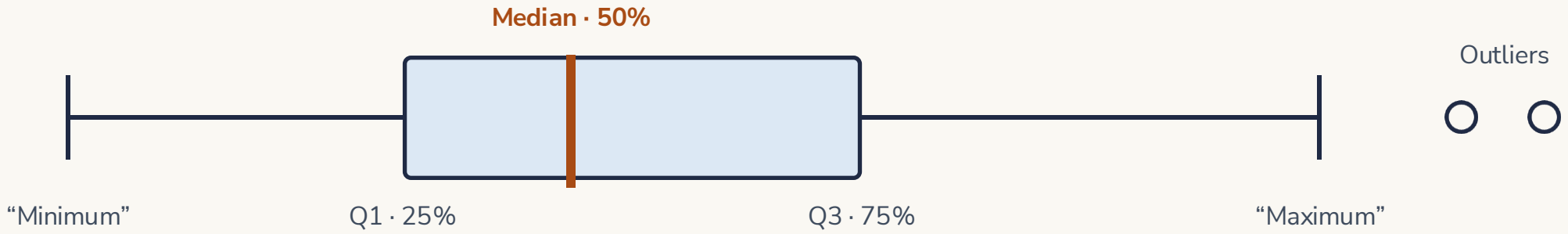
```
pyplot.hist(df2['CPU_USAGE'], bins=20)
seaborn.histplot(df2['CPU_USAGE'], bins=20)
```

Each bar counts the values in one range. **bins** sets how many bars.

Only one or two bars visible? A few huge values can squash the rest.

BOX PLOT

Reading a box plot



The box holds the middle half of the data (Q1 to Q3).

Whiskers reach at most 1.5 × the box width past the box.

```
seaborn.boxplot(
  x=df2['vsize'])
```

Compare the two apps

How many rows per app?

```
seaborn.countplot(
    x='ApplicationName', data=df2)
```

Bar height = number of rows.

Average CPU per app?

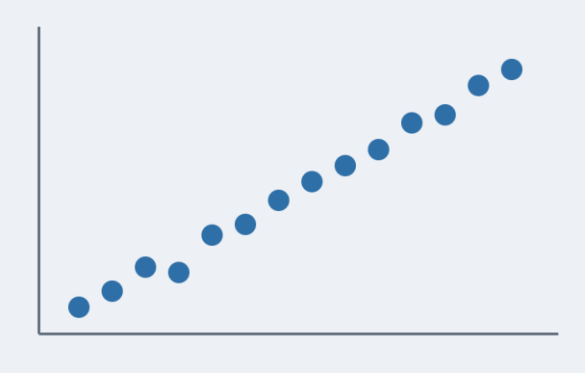
```
seaborn.barplot(x='ApplicationName',
    y='CPU_USAGE', data=df2)
```

Bar height = average. The black line shows how sure we are of it.

> Add **errorbar='sd'** to show the spread of the values (standard deviation) instead.

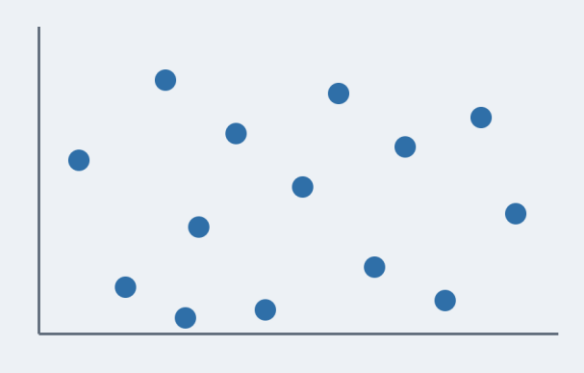
CORRELATION · SCATTER PLOT

Do two columns move together?



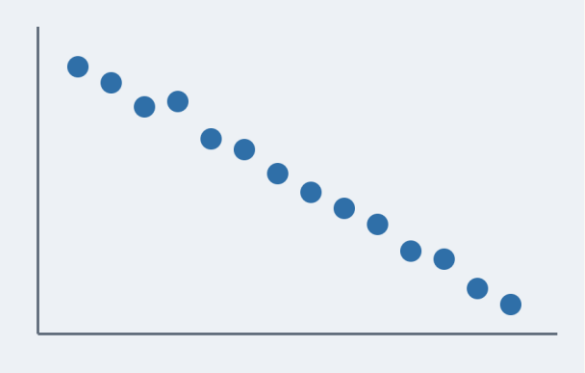
Near +1

Both go up together



Near 0

No pattern



Near −1

One up, the other down

```
seaborn.scatterplot(x='Mem', y='vsize', data=df2) # also try x='utime'
```

CORRELATION HEAT MAP

See every pair at once

```
df_corr = df2.corr(numeric_only=True) # one number per pair, from −1 to +1
pyplot.figure(figsize=(12, 12))
seaborn.heatmap(df_corr, annot=True, vmax=1, square=True, cmap='Blues')
pyplot.show()
```

Darker = stronger

annot=True also prints each number.

Diagonal is always 1

Every column matches itself.

Exactly 1 elsewhere?

Probably a duplicate column.

WRAP-UP

Write down your answers

MISSING DATA

Which column? What did we do about it?

DUPLICATE DATA

Which columns repeat other columns?

IRRELEVANT DATA

Which column added nothing?

PATTERNS

Which app uses more CPU? More memory? Which pairs correlate?

Answer these in the Conclusion cells at the end of the notebook